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(a)

Since $E(u_i|x_{i1}, x_{i2}) = 0$, $E(y_i|x_{i1}, x_{i2}) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$.

In addition, y_i is a dummy variable. Therefore,

$$\begin{aligned} E(y_i|x_{i1}, x_{i2}) &= 1 \times \Pr(y_i = 1|x_{i1}, x_{i2}) + 0 \times \Pr(y_i = 0|x_{i1}, x_{i2}) \\ &= \Pr(y_i = 1|x_{i1}, x_{i2}) \end{aligned}$$

(b) As we have learned in HE2005/HE2003, if the homoskedasticity assumption holds, $\text{Var}(u_i|x_{i1}, x_{i2}) = \sigma^2$, then $\text{Var}(y_i|x_{i1}, x_{i2}) = \sigma^2$. In other words, if $\text{Var}(y_i|x_{i1}, x_{i2})$ not constant, this implies the homoskedasticity is violated.

$y_i|x_{i1}, x_{i2}$ is a Bernoulli random variable

$$\begin{aligned} \text{Var}(y_i|x_{i1}, x_{i2}) &= \Pr(y_i = 1|x_{i1}, x_{i2}) \times [1 - \Pr(y_i = 1|x_{i1}, x_{i2})] \\ &= (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \times [1 - (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})] \end{aligned}$$

(c) Given other things being equal, if the monthly income increases by \$1000, the probability of $y = 1$ increases by 0.08 (or 7%)

Given other things being equal, if the years of education increases by 1 year, the probability of $y = 1$ increases by 0.01 (or 1%)

Note: The probability meaning takes into account the uncertainty already. We do not need to really use “on average” these two words when it comes to explaining the slope coefficients, i.e., $E(y_i|x_{i1}, x_{i2}) = Pr(y_i = 1|x_{i1}, x_{i2})$

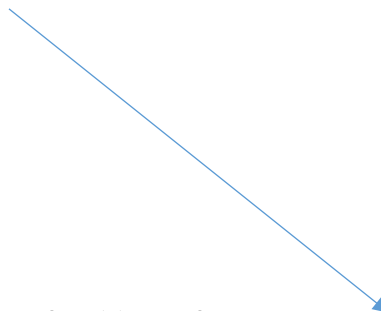
$$\begin{aligned} \text{(d) } \hat{y}_i &= 0.05 + 0.07\textit{monthly_income} + 0.01\textit{educ} \\ &= 0.05 + 0.07 \times 8 + 0.01 \times 16 = 0.77 \end{aligned}$$

$$\begin{aligned} \text{(e) } \hat{y}_i &= 0.05 + 0.07\textit{monthly_income} + 0.01\textit{educ} \\ &= 0.05 + 0.07 \times 15 + 0.01 \times 16 = 1.26 \end{aligned}$$

3.

(a) Basic Calculus theory tells us that $\frac{dF(.)}{dx} = f(.)$, i.e., a PDF is simply the derivative of a CDF.

Consider a probit model $Pr(y = 1|\mathbf{x}) = \Phi(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5)$.


$$\frac{\partial Pr(y = 1|\mathbf{x})}{\partial x_1} = \frac{\partial \Phi(.)}{\partial x_1} = \frac{\partial \Phi(.)}{\partial z} \times \frac{\partial z}{\partial x_1} = \phi(.) \times \beta_1$$

Remark: Basic Calculus theory tells us that $\frac{dF(.)}{dx} = f(.)$, i.e., a PDF is simply the derivative of a CDF.

(b)

```
. probit inlf nwifeinc educ kidslt6 age exper
```

```
Iteration 0:   log likelihood =  -514.8732
Iteration 1:   log likelihood = -407.44231
Iteration 2:   log likelihood = -406.54289
Iteration 3:   log likelihood = -406.54206
Iteration 4:   log likelihood = -406.54206
```

```
Probit regression                               Number of obs   =           753
                                                LR chi2(5)         =          216.66
                                                Prob > chi2        =           0.0000
Log likelihood = -406.54206                    Pseudo R2         =           0.2104
```

inlf	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
nwifeinc	-.0113709	.0047894	-2.37	0.018	-.020758	-.0019837
educ	.1315326	.0249731	5.27	0.000	.0825862	.180479
kidslt6	-.8862096	.1167781	-7.59	0.000	-1.11509	-.6573287
age	-.0579193	.0078292	-7.40	0.000	-.0732643	-.0425743
exper	.0691477	.0074404	9.29	0.000	.0545647	.0837307
_cons	.764545	.4400746	1.74	0.082	-.0979853	1.627075

$$Pr(\widehat{y} = 1|\mathbf{x})$$

$$= \Phi(\hat{\beta}_0 + \hat{\beta}_1nwifeinc + \hat{\beta}_2educ + \hat{\beta}_3kidslt6 + \hat{\beta}_4age + \hat{\beta}_5exper)$$

$$= \Phi(.76 - .011nwifeinc + .13educ - .89kidslt6 - .057age + .069exper)$$

$$= \Phi(0.806)$$

$$= 0.79$$

. mfx, at (nwifeinc=20 educ =10 kidslt6=0 age=30 exper =10)

Marginal effects after probit

y = Pr(inlf) (predict)

= .78997983

variable	dy/dx	Std. Err.	z	P> z	[95% C.I.]	x
nwifeinc	-.0032773	.0015	-2.18	0.029	-.006221 -.000333	20
educ	.03791	.00973	3.90	0.000	.018848 .056972	10
kidslt6	-.2554212	.03242	-7.88	0.000	-.318962 -.191881	0
age	-.0166934	.00124	-13.48	0.000	-.019121 -.014266	30
exper	.0199296	.00256	7.79	0.000	.014917 .024942	10

(c) Calculate the partial effect of $exper$ on $Pr(y = 1|x)$ for the group of women with $nwifeinc = 20$, $educ = 12$, $kidslt6 = 0$, $age = 35$, $exper = 10$ by hand.

$$\frac{\partial Pr(y = 1|x)}{\partial exper}$$

$$= \phi(\hat{\beta}_0 + \hat{\beta}_1 nwifeinc + \hat{\beta}_2 educ + \hat{\beta}_3 kidslt6 + \hat{\beta}_4 age + \hat{\beta}_5 exper) \times \hat{\beta}_5$$

(c) Calculate the partial effect of *exper* on $Pr(y = 1|x)$ for the group of women with *nwifeinc* = 20, *educ* = 12, *kidslt6* = 0, *age* = 35, *exper* = 10 by hand.

$$\frac{\partial Pr(y = 1|x)}{\partial exper}$$

$$\begin{aligned} &= \phi(\hat{\beta}_0 + \hat{\beta}_1 nwifeinc + \hat{\beta}_2 educ + \hat{\beta}_3 kidslt6 + \hat{\beta}_4 age + \hat{\beta}_5 exper) \times \hat{\beta}_5 \\ &= \phi(.76 - .011 nwifeinc + .13 educ - .89 kidslt6 - .057 age + .069 exper) \times (.069) \\ &= \phi(.76 - .011 \times 20 + .13 \times 12 - .89 \times 0 - .057 \times 35 + .069 \times 10) \times (.069) \end{aligned}$$

(c) Calculate the **partial effect** of **$exper$** on **$Pr(y = 1|x)$** for the group of women with $nwifeinc = 20$, $educ = 12$, $kidslt6 = 0$, $age = 35$, $exper = 10$ by hand.

$$\frac{\partial Pr(y = 1|x)}{\partial exper}$$

$$\begin{aligned}
 &= \phi(\hat{\beta}_0 + \hat{\beta}_1 nwifeinc + \hat{\beta}_2 educ + \hat{\beta}_3 kidslt6 + \hat{\beta}_4 age + \hat{\beta}_5 exper) \times \hat{\beta}_5 \\
 &= \phi(.76 - .011nwifeinc + .13educ - .89kidslt6 - .057age + .069exper) \times (.069) \\
 &= \phi(.76 - .011 \times 20 + .13 \times 12 - .89 \times 0 - .057 \times 35 + .069 \times 10) \times (.069) \\
 &= \phi(0.79) \times (0.069) = 0.292 \times 0.069 = 0.020
 \end{aligned}$$

```
. mfx, at(nwifeinc=20 educ=12 kidslt6=0 age=35 exper=10)
```

```

Marginal effects after probit
      y = Pr(inlf) (predict)
      = .78225156

```

variable	dy/dx	Std. Err.	z	P> z	[	95% C.I.	]	x
nwifeinc	-.003347	.00145	-2.31	0.021	-.006186	-.000508		20
educ	.0387161	.00798	4.85	0.000	.02307	.054362		12
kidslt6	-.2608526	.02795	-9.33	0.000	-.31564	-.206065		0
age	-.0170483	.00155	-11.03	0.000	-.020078	-.014019		35
exper	.0203534	.00229	8.87	0.000	.015857	.024849		10